

Linear Regression With Multiple Regressors

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Chapter Outline

- Omitted variable bias.
- Causality and regression analysis.
- Multiple regression and *OLS* estimators.
- Measures of fit.
- Sampling distribution of the *OLS* estimator.

Omitted Variable Bias

- The error u arises because of factors, or variables, that influence Y but are not included in the regression function. There are always omitted variables.
- Sometimes, the omission of those variables can lead to bias in the *OLS* estimator.
- The bias in the *OLS* estimator that occurs as a result of an omitted factor, or variable, is called *omitted variable bias*.

Omitted Variable Bias

- For omitted variable bias to occur, the omitted variable “ Z ” must satisfy two conditions
 - Z is a determinant of Y (i.e. Z is part of u); and
 - Z is correlated with the regressor X (i.e. $\text{Corr}(Z, X) \neq 0$).
- Both conditions must hold for the omission of Z to result in omitted variable bias.

Omitted Variable Bias

Class Size and Students' Performance

- English language ability (whether the student has English as a second language) plausibly affects standardized test scores: Z is a determinant of Y .
- Immigrant communities tend to be less affluent and thus have smaller school budgets and higher STR : Z is correlated with X .
- Accordingly, $\hat{\beta}_1$ is biased. What is the direction of this bias? What does common sense suggest? If common sense fails you, there is a formula...

Omitted Variable Bias

Bias Due to Omitted Variable

- If there is omitted variable bias $\rho_{Xu} \neq 0$ and the *OLS* estimator $\hat{\beta}_1$ is biased and inconsistent.

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \frac{\sigma_u}{\sigma_X} \rho_{Xu} \quad (1)$$

- For example, districts with few *ESL* students: (i) do better on standardized tests and (ii) have smaller classes (bigger budgets).
- so ignoring the effect of having many *ESL* students factor would result in overstating the class size effect. Is this actually going on in the California data?

Omitted Variable Bias

TABLE 6.1

Differences in Test Scores for California School Districts with Low and High Student–Teacher Ratios, by the Percentage of English Learners in the District

	Student–Teacher Ratio < 20		Student–Teacher Ratio ≥ 20		Difference in Test Scores, Low vs. High STR	
	Average Test Score	<i>n</i>	Average Test Score	<i>n</i>	Difference	<i>t</i> -statistic
All districts	657.4	238	650.0	182	7.4	4.04
Percentage of English learners						
< 1.9%	664.5	76	665.4	27	−0.9	−0.30
1.9–8.8%	665.2	64	661.8	44	3.3	1.13
8.8–23.0%	654.9	54	649.7	50	5.2	1.72
> 23.0%	636.7	44	634.8	61	1.9	0.68

- Districts with fewer English Learners have higher test scores.
- Districts with lower percentage of English learners (*PctEL*) have smaller classes.
- Among districts with comparable *PctEL*, the effect of class size is small (recall that overall “test score gap” = 7.4).

Omitted Variable Bias

- If an omitted variable satisfies the two conditions for omitted variable bias, then the *OLS* estimator in the regression omitting that variable is biased and inconsistent. So, even if n is large, $\hat{\beta}_1$ will not be close to β_1 .
- This raises a deeper question: how do we define β_1 ? That is, what precisely do we want to estimate when we run a regression?

Omitted Variable Bias

What precisely do we want to estimate when we run a regression? There are at least three possible answers to this question

- 1 We want to estimate the slope of a line through a scatterplot as a simple summary of the data to which we attach no substantive meaning.
- 2 We want to make forecasts, or predictions, of the value of Y for an entity not in the data set, for which we know the value of X .
- 3 We want to estimate the causal effect on Y of a change in X .

Omitted Variable Bias

Causality

In this course, we take a practical approach to defining causality: *A causal effect is defined to be the effect measured in an ideal randomized controlled experiment.*

Omitted Variable Bias

Ideal Randomized Controlled Experiments

- *Ideal*: subjects all follow the treatment protocol –perfect compliance, no errors in reporting, etc.!
- *Randomized*: subjects from the population of interest are randomly assigned to a treatment or control group (so there are no confounding factors).
- *Controlled*: having a control group permits measuring the differential effect of the treatment.
- *Experiment*: the treatment is assigned as part of the experiment: the subjects have no choice, so there is no “reverse causality” in which subjects choose the treatment they think will work best.

Omitted Variable Bias

Class Size and Students' Performance

- Imagine an ideal randomized controlled experiment for measuring the effect on Test Score of reducing STR .
- In that experiment, students would be randomly assigned to classes, which would have different sizes.
- Because they are randomly assigned, all student characteristics (and thus u_i) would be distributed independently of STR_i .
- Thus, $E[u_i|STR_i] = 0$ —that is, OLS assumption 1 holds in a randomized controlled experiment.

Omitted Variable Bias

Class Size and Students' Performance

- How does our observational data differ from this ideal? The treatment is not randomly assigned.
- Consider *PctEL* –percent English learners– in the district. It plausibly satisfies the two criteria for omitted variable bias: $Z = PctEL$ is
 - 1 a determinant of Y ; and
 - 2 correlated with the regressor X .
- Thus, the “control” and “treatment” groups differ in a systematic way, so $Corr(STR, PctEL) \neq 0$.
- Randomization + control group means that any differences between the treatment and control groups are random –not systematically related to the treatment.

Omitted Variable Bias

Class Size and Students' Performance

- We can eliminate the difference in $PctEL$ between the large (control) and small (treatment) groups by examining the effect of class size among districts with the same $PctEL$.
 - If the only systematic difference between the large and small class size groups is in $PctEL$, then we are back to the randomized controlled experiment –within each $PctEL$ group.
 - This is one way to “control” for the effect of $PctEL$ when estimating the effect of STR .

Omitted Variable Bias

There are three ways to overcome the omitted variable bias

- 1 Run a randomized controlled experiment in which treatment (STR) is randomly assigned: then $PctEL$ is still a determinant of $TestScore$, but uncorrelated with STR (rarely feasible solution).
- 2 Adopt the “cross tabulation” approach, with finer gradations of STR and $PctEL$ –within each group, all classes have the same $PctEL$, so we control for $PctEL$ (But soon you will run out of data, and what about other determinants like family income and parental education?)
- 3 Use a regression in which the omitted variable ($PctEL$) is no longer omitted: include $PctEL$ as an additional regressor in a multiple regression.

The Population Multiple Regression Model

Linear Regression Model With Multiple Regressors

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_K X_{Ki} + u_i, \quad \forall i = 1, \dots, n. \quad (2)$$

- We have n observations $(Y_i, X_{1i}, \dots, X_{Ki})$.
- $X_1, \dots, X_K$ are the *independent* variables and Y is the *dependent* variable.
- β_0 is the intercept and $\beta_1, \dots, \beta_K$ are the slope parameters.
- u_i is the regression error, and it consists of omitted factors and errors in measurement of Y .

The Population Multiple Regression Model

Interpretation of the Estimated Slope Coefficients

Write the population model as

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_K X_K + u. \quad (3)$$

Consider changing X_1 by ΔX_1 while holding X_k constant $\forall k = 2, \dots, K$.
We can show

$$\beta_1 = \frac{\Delta Y}{\Delta X_1}, \text{ holding } X_k \text{ constant } \forall k = 2, \dots, K. \quad (4)$$

In general,

$$\beta_s = \frac{\Delta Y}{\Delta X_s}, \text{ holding } X_k \text{ constant } \forall k \neq s; k = 1, \dots, K. \quad (5)$$

OLS Estimator in The Multiple Regression Model

The OLS Estimator

Let $b_0, b_1, \dots, b_K$ be some estimators of $\beta_0, \beta_1, \dots, \beta_K$. Then, the predicted value of Y_i using this line is

$$\hat{Y}_i = b_0 + b_1 X_{1i} + \dots + b_K X_{Ki}.$$

Thus, the mistake made in predicting the i -th observation is

$$\begin{aligned}\hat{u}_i &= Y_i - \hat{Y}_i = Y_i - (b_0 + b_1 X_i + \dots + b_K X_{Ki}) \\ &= Y_i - b_0 - b_1 X_i - \dots - b_K X_{Ki},\end{aligned}$$

and the sum of the squared mistakes is

$$\sum_{i=1}^n (Y_i - b_0 - b_1 X_i - \dots - b_K X_{Ki})^2.$$

OLS Estimator in The Multiple Regression Model

The OLS Estimator

The *OLS* estimators of $\beta_0, \beta_1, \dots, \beta_K$ are the expressions for $b_0, b_1, \dots, b_K$ minimizing the sum of squared errors above. We will denote these estimators by $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_K$.

Case Study

Class Size and Students' Performance: The California Data

Our previous regression showed

$$\widehat{TestScore} = 698 - 2.28 \times STR.$$

Now, including the percentage of English learners ($PctEL$) gives

$$\widehat{TestScore} = 686.0 - 1.1 \times STR - 0.65 \times PctEL.$$

What happened to the coefficient of STR ? Why? (Note that $Corr(STR, PctEL) = 0.19$).

OLS Regression: *STATA* Output

```
reg testscr str pctel, robust;
```

Regression with robust standard errors

Number of obs = 420
F(2, 417) = 223.82
Prob > F = 0.0000
R-squared = 0.4264
Root MSE = 14.464

		Robust					
testscr		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+							
str		-1.101296	.4328472	-2.54	0.011	-1.95213	-.2504616
pctel		-.6497768	.0310318	-20.94	0.000	-.710775	-.5887786
_cons		686.0322	8.728224	78.60	0.000	668.8754	703.189

Measures of Fit for Multiple Regression

Recall

- $Y_i = \hat{Y}_i + \hat{u}_i$: actual = predicted + residual.
- SER : standard deviation of $\hat{u}_i$ (with d.f. correction).
- $RMSE$: standard deviation of $\hat{u}_i$ (without d.f. correction).
- R^2 : fraction of variance of Y explained by X .
- $\bar{R}^2$: Adjusted R^2 is the R^2 with d.f. correction, $\bar{R}^2 < R^2$.

Measures of Fit for Multiple Regression

SER and *RMSE*

- As in regression with a single regressor, the *SER* and the *RMSE* are measures of the spread of the Y 's around the regression line

$$SER = s_{\hat{u}} = \sqrt{\frac{1}{n-K-1} \sum_{i=1}^n (\hat{u}_i - \bar{\hat{u}})^2} = \sqrt{\frac{1}{n-K-1} \sum_{i=1}^n \hat{u}_i^2}.$$

- The second equality comes from the fact that $\bar{\hat{u}} = \frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0$.
- The *RMSE* is defined as before

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n \hat{u}_i^2}. \quad (6)$$

Measures of Fit for Multiple Regression

R^2 and Adjusted R^2

- The R^2 is the fraction of the variance explained –same definition as in regression with a single regressor

$$R^2 = \frac{ESS}{TSS}. \quad (7)$$

- From the definition of predicted value we can write

$$ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2, \quad TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2, \quad SSR = \sum_{i=1}^n \hat{u}_i^2. \quad (8)$$

- We can also write the R^2 as

$$R^2 = 1 - \frac{SSR}{TSS}. \quad (9)$$

Measures of Fit for Multiple Regression

R^2 and Adjusted R^2

- The R^2 always increases when you add another regressor (why?) –a bit of a problem for a measure of “fit”.
- The “adjusted R^2 ” or $\bar{R}^2$ corrects this problem by “penalizing” you for including another regressor –the adjusted R^2 does not necessarily increase when you add another regressor.

$$\bar{R}^2 = 1 - \left(\frac{n-1}{n-K-1} \right) \frac{SSR}{TSS}. \quad (10)$$

- Note that $\bar{R}^2 < R^2$ but their value is very close when n is large.

Case Study

- Estimating the model in chapter 4 gives

$$\widehat{TestScore} = 698.9 - 2.28 \times STR$$

$$R^2 = 0.05, \text{ SER} = 18.6$$

- Estimating the model in chapter 6 gives

$$\widehat{TestScore} = 686 - 1.1 \times STR - 0.65PctEL$$

$$R^2 = 0.426, \bar{R}^2 = 0.424, \text{ SER} = 14.5$$

- What does this tell you about the fit of regression in chapter 6 compared with regression in chapter 4? Why are the R^2 and $\bar{R}^2$ so close?

OLS Assumptions for Multiple Regression

Recall our population model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_K X_{Ki} + u_i, \quad \forall i = 1, \dots, n.$$

Least Square Assumptions

- 1 **The conditional distribution of u_i given the X 's has mean zero.**
That is, $E[u_i | X_{1i} = x_1, \dots, X_{Ki} = x_K] = 0, \quad \forall i = 1, \dots, n.$
- 2 **$(X_{1i}, \dots, X_{Ki}, Y_i), i = 1, \dots, n,$ are i.i.d.**
- 3 **Large outliers in X and/or Y are rare.** Technically,
 $0 < E[X_{1i}^4] < \infty, \dots, E[X_{Ki}^4] < \infty,$ and $0 < E[Y_i^4] < \infty$
- 4 **There is no perfect multicollinearity.**

Least Squares Assumptions

Assumption 1: $E[u|X_1, \dots, X_K] = 0$

- This has the same interpretation as in regression with a single regressor.
- Failure of this condition leads to omitted variable bias, specifically, if an omitted variable belongs in the equation (so is in u) and is correlated with an included X then this condition fails and there is OV bias.
- The best solution, if possible, is to include the omitted variable in the regression.
- A second, related solution is to include a variable that controls for the omitted variable (discussed in Ch. 7)

Least Squares Assumptions

Assumption 2: $(X_{1i}, \dots, X_{Ki}, Y_i)$, $i = 1, \dots, n$, are i.i.d.

- This is satisfied automatically if the data are collected by simple random sampling.

Assumption 3: large outliers are rare (finite fourth moments)

- As in the case of a single regressor, *OLS* can be sensitive to large outliers, so you need to check your data (scatterplots!) to make sure there are no crazy values (typos or coding errors).

Least Squares Assumptions

Assumption 4: There is no perfect multicollinearity

Perfect multicollinearity occurs when one of the regressors is an exact linear function of other regressors. This happens, for example, when you accidentally include the same variable twice.

Sampling Distribution of the *OLS* Estimator

Under the four *OLS* assumptions above

- The sampling distribution of $\hat{\beta}_k$ has mean β_k .
- $\text{Var}[\hat{\beta}_k]$ is inversely proportional to n .
- Other than mean and variance the exact (finite sample) distribution of $\hat{\beta}_k$ is very complicated.
- For large n we can rely on the asymptotic approximation
 - $\hat{\beta}_k$ is consistent: $\hat{\beta}_k \xrightarrow{P} \beta_k$.
 - $\frac{\hat{\beta}_k - E[\hat{\beta}_k]}{\sqrt{\text{Var}[\hat{\beta}_k]}}$ is approximately distributed as a $N(0, 1)$ variable (by the *CLT*)
 - These statements hold for all variable k .

Perfect and Imperfect Multicollinearity

Perfect multicollinearity

Perfect multicollinearity is when one of the regressors is an exact linear function of the other regressors. In our *test score regression* some examples include

- you include *STR* twice,
- you regress *TestScore* on two constants, D_i and B_i , where: $D_i = 1$ if $STR \leq 20$ and $D_i = 0$ otherwise; while $B_i = 1$ if $STR > 20$ and $B_i = 0$ otherwise. Then $B_i = 1 - D_i$ and there is perfect multicollinearity.

Perfect and Imperfect Multicollinearity

The Dummy Variable Trap

- Suppose you have a set of multiple binary (dummy) variables, which are mutually exclusive and exhaustive –that is, there are multiple categories and every observation falls in one and only one category (Freshmen, Sophomores, Juniors, and Seniors).
- If you include all these dummy variables and a constant, you will have perfect multicollinearity –this called the *dummy variable trap*.
- Why is there perfect multicollinearity here?
- Solutions to the dummy variable trap: (1) omit one of the groups (e.g. Senior), or (2) omit the intercept.
- By convention, we keep the constant and omit one of the groups. What are the implications for the interpretation of the coefficients?

Perfect and Imperfect Multicollinearity

Perfect Multicollinearity

- Perfect multicollinearity usually reflects a mistake in the definitions of the regressors, or an oddity in the data.
- Your statistical software will let you know –either by crashing and giving an error message or by “dropping” one of the variables arbitrarily.
- The solution to perfect multicollinearity is to modify your list of regressors so that you no longer have perfect multicollinearity.

Perfect and Imperfect Multicollinearity

Imperfect Multicollinearity

Imperfect multicollinearity occurs when two or more regressors are very highly correlated.

- Imperfect and perfect multicollinearity are quite different despite the similarity of the names.
- Why the term “multicollinearity”? If two regressors are very highly correlated, then their scatterplot will pretty much look like a straight line –they are “colinear” –but unless the correlation is exactly 1 or -1 , that collinearity is imperfect.
- Imperfect multicollinearity implies that one or more of the regression coefficients will be imprecisely estimated.

Perfect and Imperfect Multicollinearity

Imperfect Multicollinearity

- The coefficient on X_1 is the effect of X_1 holding every other X constant; but if X_1 and X_2 are highly correlated, there is very little variation in X_1 once X_2 is held constant –so the data does not contain much information about what happens when X_1 changes but X_2 does not.
- If so, the variance of the *OLS* estimator of the coefficient on X_1 will be large.
- Imperfect multicollinearity (correctly) results in large standard errors for one or more of the *OLS* coefficients.